

Bank Management System - C++ Source Code

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#include <iostream>
#include <fstream>
#include <string>
#include <iomanip>

using namespace std;

class BankAccount {
private:
    string accountNumber;
    string accountHolderName;
    double balance;

public:
    // Constructor
    BankAccount() : accountNumber(""), accountHolderName(""), balance(0.0) {}

    // Function to create a new account
    void createAccount() {
        cout << "\n--- Create New Account ---" << endl;
        cout << "Enter Account Number: ";
        cin >> accountNumber;
        cin.ignore(); // Clear newline character from buffer
        cout << "Enter Account Holder Name: ";
        getline(cin, accountHolderName);
        cout << "Enter Initial Deposit ($): ";
        cin >> balance;
        cout << "\nAccount successfully created!" << endl;
    }

    // Function to display account details
    void displayAccount() const {
        cout << "\n-----" << endl;
        cout << "Account Details" << endl;
        cout << "-----" << endl;
        cout << "Account Number : " << accountNumber << endl;
        cout << "Holder Name      : " << accountHolderName << endl;
        cout << "Current Balance: $" << fixed << setprecision(2) << balance <<
endl;
    }

    // Function to deposit money
    void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            cout << "\n$" << amount << " deposited successfully!" << endl;
        } else {
            cout << "\nInvalid deposit amount." << endl;
        }
    }

    // Function to withdraw money
    void withdraw(double amount) {
        if (amount > balance) {
            cout << "\nInsufficient balance! Transaction failed." << endl;
        } else if (amount <= 0) {
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        cout << "\nInvalid withdrawal amount." << endl;
    } else {
        balance -= amount;
        cout << "\n$" << amount << " withdrawn successfully!" << endl;
    }
}

// Getter for account number to help with searching
string getAccountNumber() const {
    return accountNumber;
}
};

// Main function managing the menu system
int main() {
    BankAccount account;
    int choice;
    double amount;
    bool accountExists = false;

    do {
        cout << "\n===== " << endl;
        cout << "      BANK MANAGEMENT SYSTEM      " << endl;
        cout << "===== " << endl;
        cout << "1. Create New Account" << endl;
        cout << "2. Deposit Money" << endl;
        cout << "3. Withdraw Money" << endl;
        cout << "4. Balance Inquiry" << endl;
        cout << "5. Exit" << endl;
        cout << "===== " << endl;
        cout << "Enter your choice (1-5): ";
        cin >> choice;

        switch (choice) {
            case 1:
                account.createAccount();
                accountExists = true;
                break;

            case 2:
                if (accountExists) {
                    cout << "Enter amount to deposit ($): ";
                    cin >> amount;
                    account.deposit(amount);
                } else {
                    cout << "\nError: No account found. Please create an account
first." << endl;
                }
                break;

            case 3:
                if (accountExists) {
                    cout << "Enter amount to withdraw ($): ";
                    cin >> amount;
                    account.withdraw(amount);
                } else {
                    cout << "\nError: No account found. Please create an account
first." << endl;
                }
                break;

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        case 4:
            if (accountExists) {
                account.displayAccount();
            } else {
                cout << "\nError: No account found. Please create an account
first." << endl;
            }
            break;

        case 5:
            cout << "\nThank you for using our Banking System. Goodbye!" <<
endl;
            break;

        default:
            cout << "\nInvalid choice! Please select a valid option." <<
endl;
    }
} while (choice != 5);

return 0;
}
```